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"""Generate a SAMPLE Papoine orders dataset (not real business figures).

Run:  python generate_sample_data.py
Creates papoine_orders_sample.csv
"""
import random
from datetime import date, timedelta

import pandas as pd

random.seed(20261014)

BOUQUETS = {  # base price EUR, seasonal weight per month (Jan..Dec)
    "Riga Rose Classic": (38, [1, 3, 1.2, 1, 1, .9, .9, .9, 1, 1, 1, 1.3]),
    "Tulip Spring": (29, [1.5, 1.8, 3, 3, 2.5, .3, .3, .3, .3, .3, .3, .5]),
    "Peony Blush": (45, [.4, .4, .4, .4, 3, 3, 2.5, .4, .4, .4, .4, .4]),
    "Sunflower Glow": (27, [.3, .3, .3, .3, .3, 1, 2.5, 2.5, 2.5, .6, .3, .3]),
    "Wild Meadow": (32, [.6, .6, .6, .8, 1.6, 1.6, 1.6, 1.6, 1.6, .8, .6, .6]),
    "Seasonal Surprise": (35, [1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1.2, 2]),
}
PLANS = {"One-off": 45, "Weekly": 12, "Bi-weekly": 18, "Monthly": 25}
PLAN_FREQ = {"One-off": 0.6, "Weekly": 4, "Bi-weekly": 2, "Monthly": 1}
AREAS = {"Centrs": 25, "Āgenskalns": 15, "Teika": 14, "Purvciems": 16,
         "Jūrmala": 12, "Mežaparks": 8, "Imanta": 10}
CHANNELS = {"Website": 45, "Instagram": 30, "Referral": 15, "Gift voucher": 10}
SIZES = {"Small": (30, 0.75), "Medium": (45, 1.0), "Large": (25, 1.4)}


def pick(weights: dict):
    return random.choices(list(weights), weights=list(weights.values()))[0]


customers = []
for i in range(1, 261):
    plan = pick(PLANS)
    customers.append((f"C{i:03d}", plan))
cust_weights = [PLAN_FREQ[p] for _, p in customers]

start, end = date(2025, 11, 1), date(2026, 10, 13)
total_days = (end - start).days
rows, n = [], 1

for d in range(total_days + 1):
    day = start + timedelta(days=d)
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    m, dom, wd = day.month, day.day, day.weekday()  # Monday=0
    lam = 1.4 + 0.8 * d / total_days
    if m == 2 and 10 <= dom <= 14: lam *= 4      # Valentine's Day
    if m == 3 and 6 <= dom <= 8: lam *= 5        # Women's Day
    if m == 5 and 8 <= dom <= 10: lam *= 3       # Mother's Day
    if m == 12 and 18 <= dom <= 24: lam *= 2.5   # Christmas
    if wd in (4, 5): lam *= 1.3
    if wd == 6: lam *= 0.7

    for _ in range(round(lam * (0.6 + random.random() * 0.8))):
        cid, plan = random.choices(customers, weights=cust_weights)[0]
        bouquet = random.choices(list(BOUQUETS), weights=[b[1][m - 1] for b in BO
        size = random.choices(list(SIZES), weights=[s[0] for s in SIZES.values()]
        discount = 1 if plan == "One-off" else 0.9
        price = round(BOUQUETS[bouquet][0] * SIZES[size][1] * discount * 2) / 2
        cancel = 9 if plan == "Weekly" else 5
        status = pick({"Delivered": 100 - cancel - 3, "Cancelled": cancel, "Refun
        rating = (pick({5: 50, 4: 30, 3: 12, 2: 5, 1: 3})
                  if status == "Delivered" and random.random() < 0.55 else None)
        rows.append({
            "order_id": f"P-{n:05d}", "order_date": day.isoformat(), "month": day
            "weekday": day.strftime("%A"), "customer_id": cid, "plan_type": plan,
            "bouquet": bouquet, "size": size, "price_eur": price,
            "delivery_area": pick(AREAS), "channel": pick(CHANNELS),
            "status": status, "rating": rating,
        })
        n += 1

df = pd.DataFrame(rows)
df["rating"] = df["rating"].astype("Int64")
df.to_csv("papoine_orders_sample.csv", index=False)
print(f"Saved {len(df)} rows to papoine_orders_sample.csv")
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